Cooper, Gutierrez, and Marcum argued implicitly that since true value indexes weren't created until after 1992, when Eugene Fama and Ken French came out with their now groundbreaking paper entitled "The Cross-Section of Expected Stock Returns" (*Journal of Finance,* 1992), the managers are being unduly criticized. In this example, the money manager would have no reason in 1980 to think that low price-to-book ratio stocks would outperform the market because that was not a recognized strategy. Thus, this manager's track record should be compared to a value index only from 1992 to 2005 rather than from 1980 to 2005.

An analogy can be found in the sport of baseball. The concept of "relief pitching" did not appear until after the turn of the twentieth century. To make a point, assume that prior to 1900 pitchers only pitched complete games and that after 1900 relief pitching came into being. Also assume that immediately after relief pitching began, the idea of "earned run average" (ERA) began to be used (in actuality, there were some years in between these events, but this scenario approximates reality). For those unfamiliar with baseball, ERA is the number of earned runs a pitcher gives up per nine innings pitched.

Prior to 1900, pitchers were routinely expected to pitch complete games. Thus, the only way to measure their effectiveness was to look at their win-loss (W-L) record. After the ERA was introduced, pitchers were evaluated on both W-L and ERA. Suppose that a statistician went back and calculated the ERA for pitchers who pitched prior to 1900 and found that their ERAs were much higher (bad) than those who were evaluated after relief pitching was introduced. Evaluating the pre-1900 pitchers by ERA is unfair because ERA was not known prior to 1900 as a measure of effectiveness of a pitcher, as pitchers were forced to pitch complete games. This situation is analogous to the previous example of value managers prior to 1992.

From their paper, Cooper, Gutierrez, and Marcum noted:

> *In this study, we address the* ex ante *predictability of the cross section of stock returns by investigating whether a real-time investor could have used book-to-market equity, firm size, and one-year lagged returns to generate portfolio profits during the 1974–97 period. We develop variations on common recursive out-of-sample methods and demonstrate a marked difference between* ex post *and* ex ante *predictability, suggesting that the current notion of predictability in the literature is exaggerated.*[4]
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> —(There is much more to this article. At this writing, it can be found at http://lcb1.uoregon.edu/rcg/research/Realtime.pdf.)

A similar example lies with small-capitalization stocks. These managers should be judged from the early 1980s forward, when the idea of